Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division C | Protein Modeling | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Alpha-helix H bonds mainly involve…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Only neighboring side-chain colors

B. Backbone groups

C. Only sulfur atoms

D. Only DNA bases

Correct: B

2. (2 points) Attachment and fusion are…

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only DNA processes

B. Absent in all enveloped viruses

C. Different entry-related steps

D. Always identical

Correct: C

3. (2 points) Rotation changes…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain's helix region.

A. Primary sequence

B. Chemical bonds automatically

C. PDB identifier

D. Viewpoint

Correct: D

4. (2 points) Quaternary structure requires…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. No interfaces

B. Multiple polypeptide subunits

C. Exactly four atoms

D. Only one residue

Correct: B

5. (2 points) An epitope is…

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Every viral genome

B. A unit of salinity

C. A recognized molecular feature

D. The entire immune system

Correct: C

6. (2 points) Color by structure shows…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

A. Measured free energy

B. Every atom’s charge

C. A diagnosis

D. A structural classification

Correct: D

7. (2 points) Denaturation always cleaves peptide bonds?

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Only at neutral pH

B. No

C. Always

D. Only for helices

Correct: B

8. (2 points) Binding alone proves neutralization?

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only for any antibody

B. Only in a diagram

C. No

D. Always

Correct: C

9. (2 points) The built-in lab executes full Jmol?

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

A. Yes

B. Only at maximum zoom

C. Only for HA

D. No

Correct: D

10. (4 points) Compare motif and domain.

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

answer: A motif is a recurring arrangement; a domain is a larger structural/functional unit. (4 points).

11. (2 points) The assigned interval runs from residue 153 through 252, inclusive. How many numbered residue positions are included?

answer: 100 (2 points).

Numerical target 100; absolute tolerance 0.2. Accepted units: , .

12. (2 points) At the specified scale of 2 cm per residue, what backbone material length corresponds to 100 residues before folding?

answer: 200 cm (2 points).

Numerical target 200; absolute tolerance 0.4. Accepted units: , cm.

13. (2 points) State the assigned 2027 structure identifier, chain and residue interval, and distinguish these identifiers from a claim that all residues are experimentally resolved.

answer: PDB 3UBE, chain A, residues 153-252. A numbered interval specifies the requested region; experimental missing residues must be checked in the structure record. (2 points).

14. (2 points) Explain the functional relevance of hemagglutinin binding to sialic-acid receptors in influenza entry.

answer: Binding helps the virus attach to susceptible host-cell receptors and initiates the entry pathway; binding specificity is related to infection biology. This is State/National molecular detail in the supplied rules. (2 points).

15. (2 points) Why does identifying an alpha helix require backbone geometry rather than simply finding a hydrophobic sequence stretch?

answer: Secondary structure depends on backbone conformation and hydrogen-bond organization; residue composition alone does not uniquely establish the actual geometry. (2 points).

16. (2 points) A synthetic structure worksheet places two oppositely charged side chains close together. Describe a possible stabilizing interaction and a qualification.

answer: An electrostatic/salt-bridge interaction may stabilize the arrangement, but protonation state, solvent, distance and geometry affect its strength. (2 points).

17. (2 points) Explain why displaying a ligand-binding pocket without identifying chain and residue numbering can mislead a model reviewer.

answer: Equivalent residue numbers can refer to different chains or numbering schemes; reproducible selection requires structure, chain, residue identifiers and consistent coordinate context. (2 points).

18. (2 points) What is the distinction between a displayed molecular contact and a proven functional interaction?

answer: A geometrical contact is structural evidence; functional significance requires appropriate chemical context and supporting evidence rather than proximity alone. (2 points).

19. (2 points) A model includes a decorative feature unrelated to protein structure or function. Explain why it does not improve its scientific explanation.

answer: Useful features should communicate relevant structural or functional evidence and be identified in the legend; irrelevant decoration does not explain the molecule. (2 points).

20. (2 points) Explain why rotating a JUDE molecular view does not alter the primary sequence or chemical connectivity.

answer: Rotation changes the viewing transformation, not residue order or the underlying molecular bond structure. (2 points).